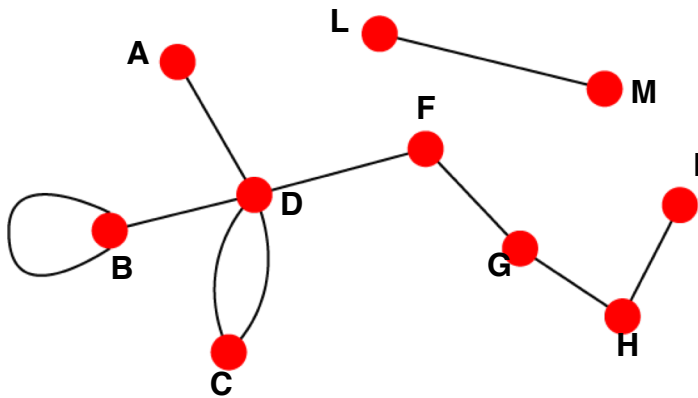


Choice of Network Representation

Directed vs. Undirected Graphs

Undirected

- **Links:** undirected (symmetrical, reciprocal)

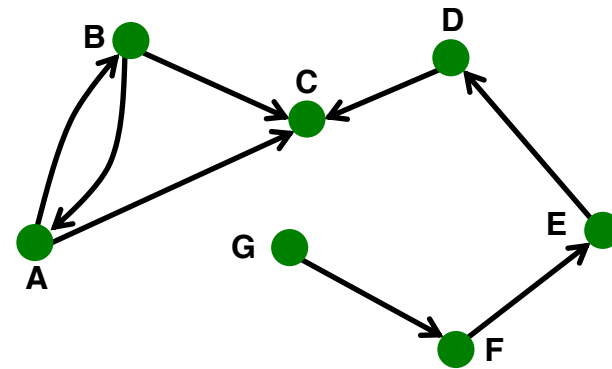


- **Examples:**

- Collaborations
- Friendship on Facebook

Directed

- **Links:** directed (arcs)

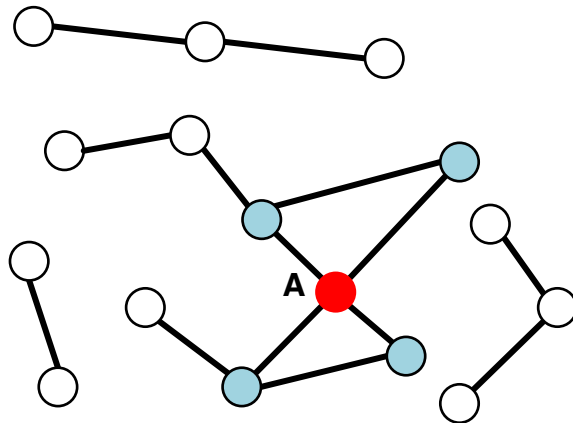


- **Examples:**

- Phone calls
- Following on Twitter

Node Degrees

Undirected

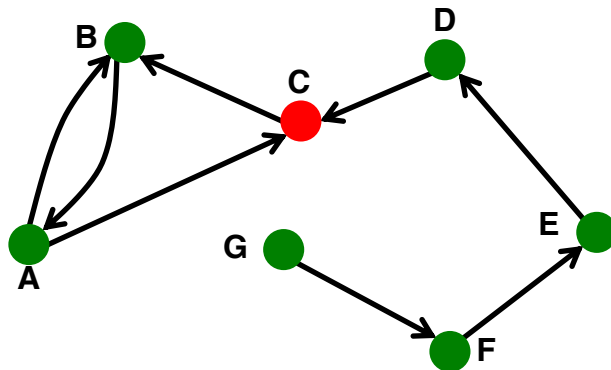


Node degree, k_i : the number of edges adjacent to node i

$$k_A = 4$$

Avg. degree: $\bar{k} = \langle k \rangle = \frac{1}{N} \sum_{i=1}^N k_i = \frac{2E}{N}$

Directed



In directed networks we define an **in-degree** and **out-degree**.

The (total) degree of a node is the sum of in- and out-degrees.

$$k_C^{in} = 2 \quad k_C^{out} = 1 \quad k_C = 3$$

$$\bar{k} = \frac{E}{N}$$

$$\overline{k^{in}} = \overline{k^{out}}$$

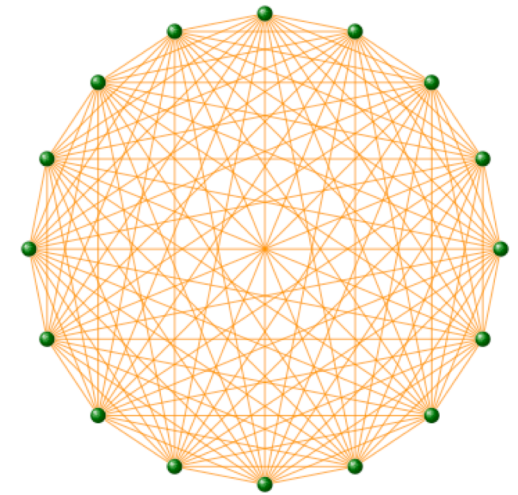
Source: Node with $k^{in} = 0$

Sink: Node with $k^{out} = 0$

Complete Graph

The **maximum number of edges** in an undirected graph on N nodes is

$$E_{\max} = \binom{N}{2} = \frac{N(N-1)}{2}$$



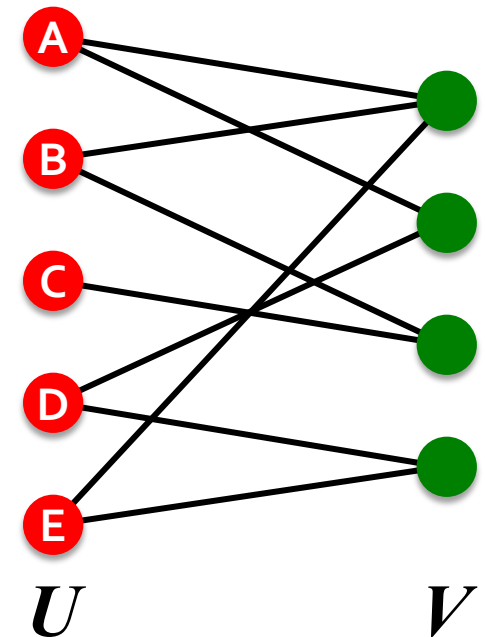
An undirected graph with the number of edges $E = E_{\max}$ is called a **complete graph**, and its average degree is $N-1$

Directedness & Average Degrees

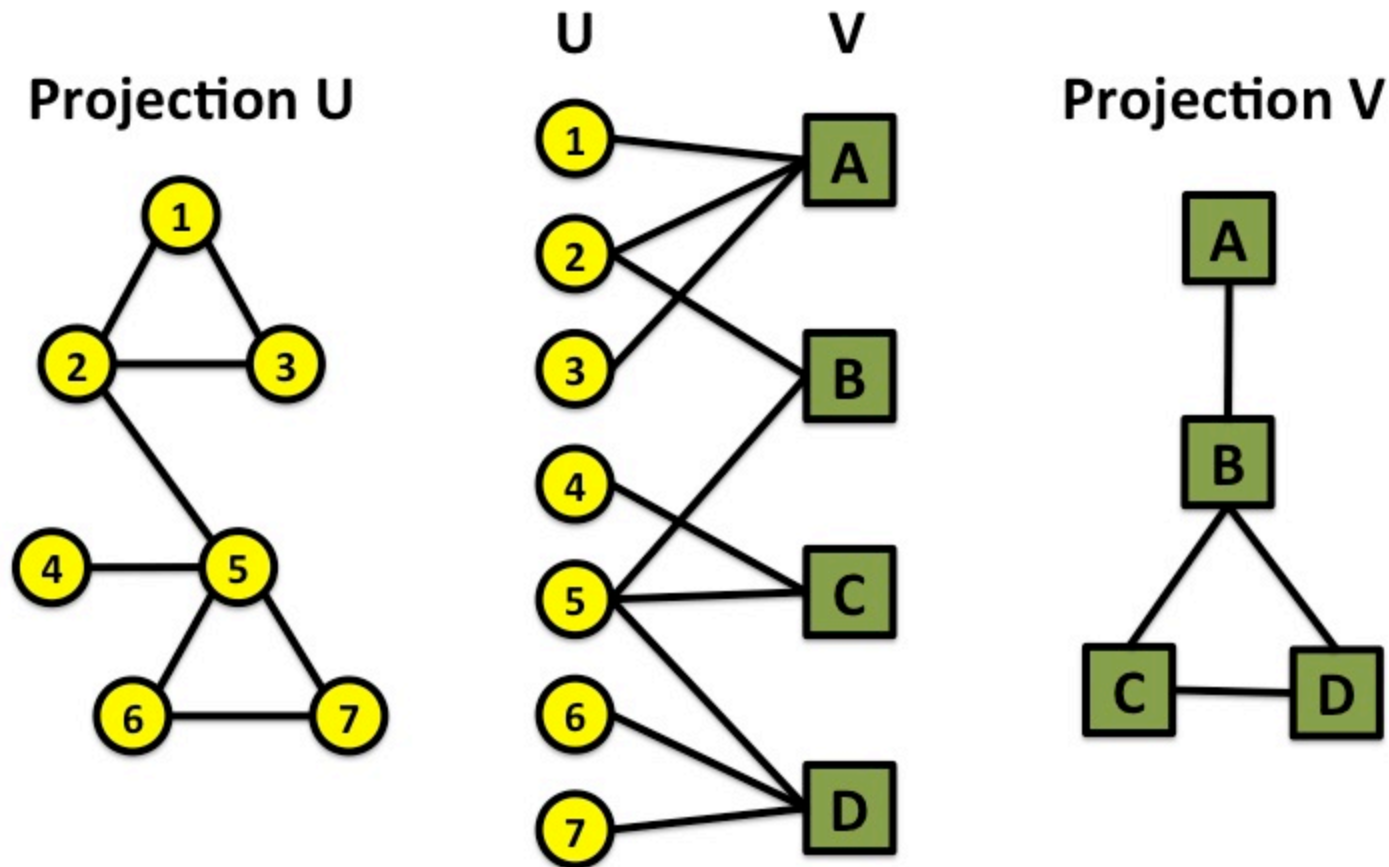
NETWORK	NODES	LINKS	DIRECTED UNDIRECTED	N	L	$\langle k \rangle$
Internet	Routers	Internet connections	Undirected	192,244	609,066	6.33
WWW	Webpages	Links	Directed	325,729	1,497,134	4.60
Power Grid	Power plants, transformers	Cables	Undirected	4,941	6,594	2.67
Mobile Phone Calls	Subscribers	Calls	Directed	36,595	91,826	2.51
Email	Email addresses	Emails	Directed	57,194	103,731	1.81
Science Collaboration	Scientists	Co-authorship	Undirected	23,133	93,439	8.08
Actor Network	Actors	Co-acting	Undirected	702,388	29,397,908	83.71
Citation Network	Paper	Citations	Directed	449,673	4,689,479	10.43
E. Coli Metabolism	Metabolites	Chemical reactions	Directed	1,039	5,802	5.58
Protein Interactions	Proteins	Binding interactions	Undirected	2,018	2,930	2.90

Bipartite Graph

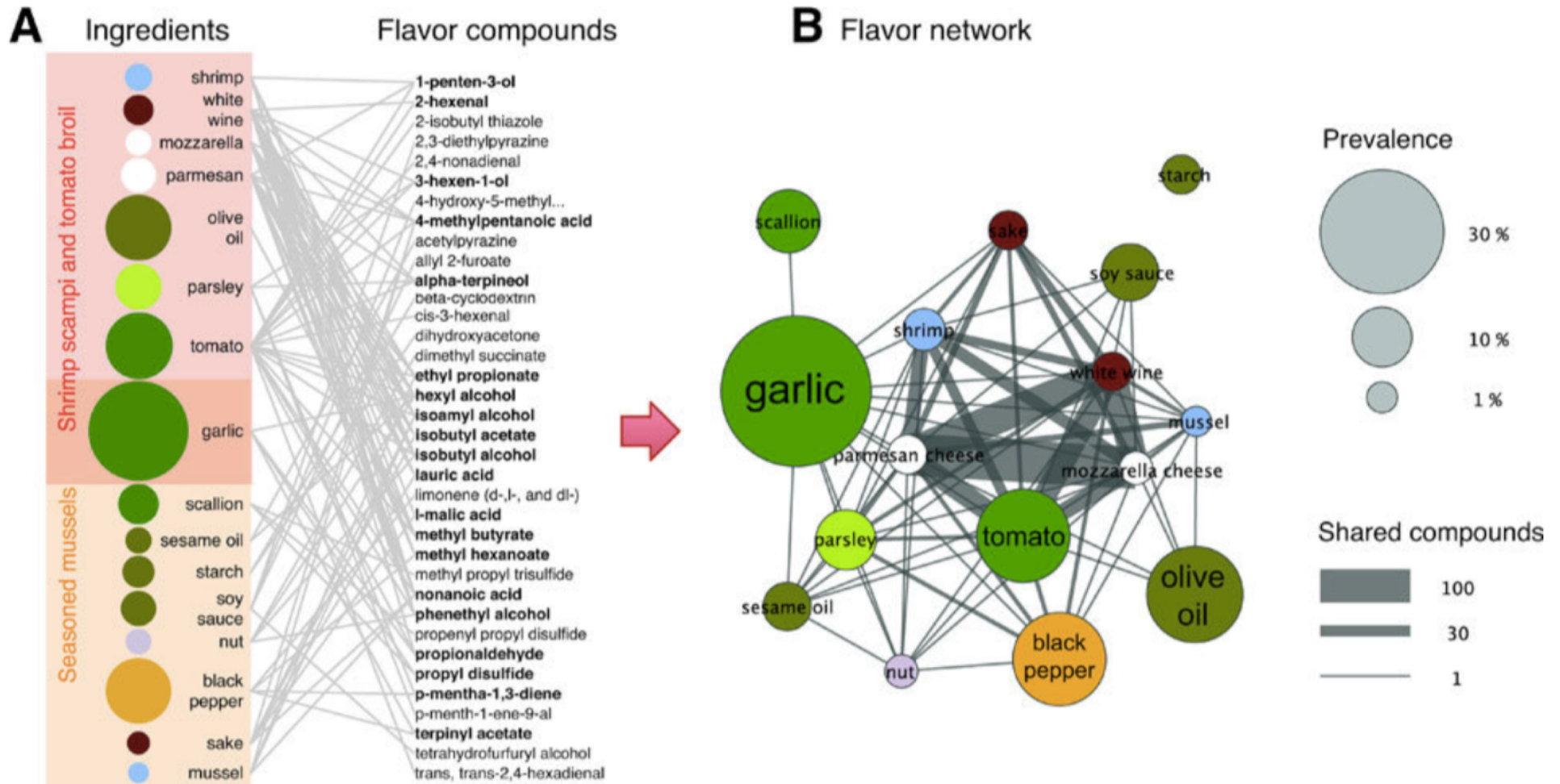
- **Bipartite graph** is a graph whose nodes can be divided into two disjoint sets U and V such that every link connects a node in U to one in V ; that is, U and V are **independent sets**
- **Examples:**
 - Authors-to-Papers (they authored)
 - Actors-to-Movies (they appeared in)
 - Users-to-Movies (they rated)
 - Recipes-to-Ingredients (they contain)
- **“Folded” networks:**
 - Author collaboration networks
 - Movie co-rating networks



Folded/Projected Bipartite Graphs



Example: Ingredients and Flavors

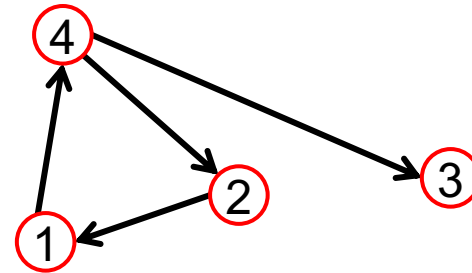
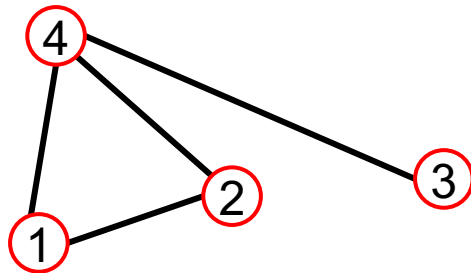


Y.-Y. Ahn, S. E. Ahnert, J. P. Bagrow, A.-L. Barabási

Flavor network and the principles of food pairing , *Scientific Reports* 196, (2011).

Network Science: Graph Theory

Representing Graphs: Adjacency Matrix



$A_{ij} = 1$ if there is a link from node i to node j
 $A_{ij} = 0$ otherwise

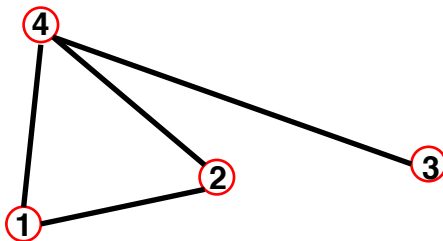
$$A = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

$$A = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

Note that for a directed graph (right) the matrix is not symmetric.

Adjacency Matrix

Undirected



$$A_{ij} = \begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

$$A_{ij} = A_{ji}$$

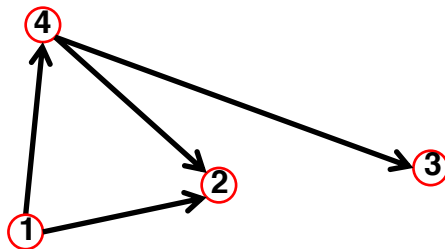
$$A_{ii} = 0$$

$$k_i = \sum_{j=1}^N A_{ij}$$

$$k_j = \sum_{i=1}^N A_{ij}$$

$$L = \frac{1}{2} \sum_{i=1}^N k_i = \frac{1}{2} \sum_{ij} A_{ij}$$

Directed



$$A = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}$$

$$A_{ij} \neq A_{ji}$$

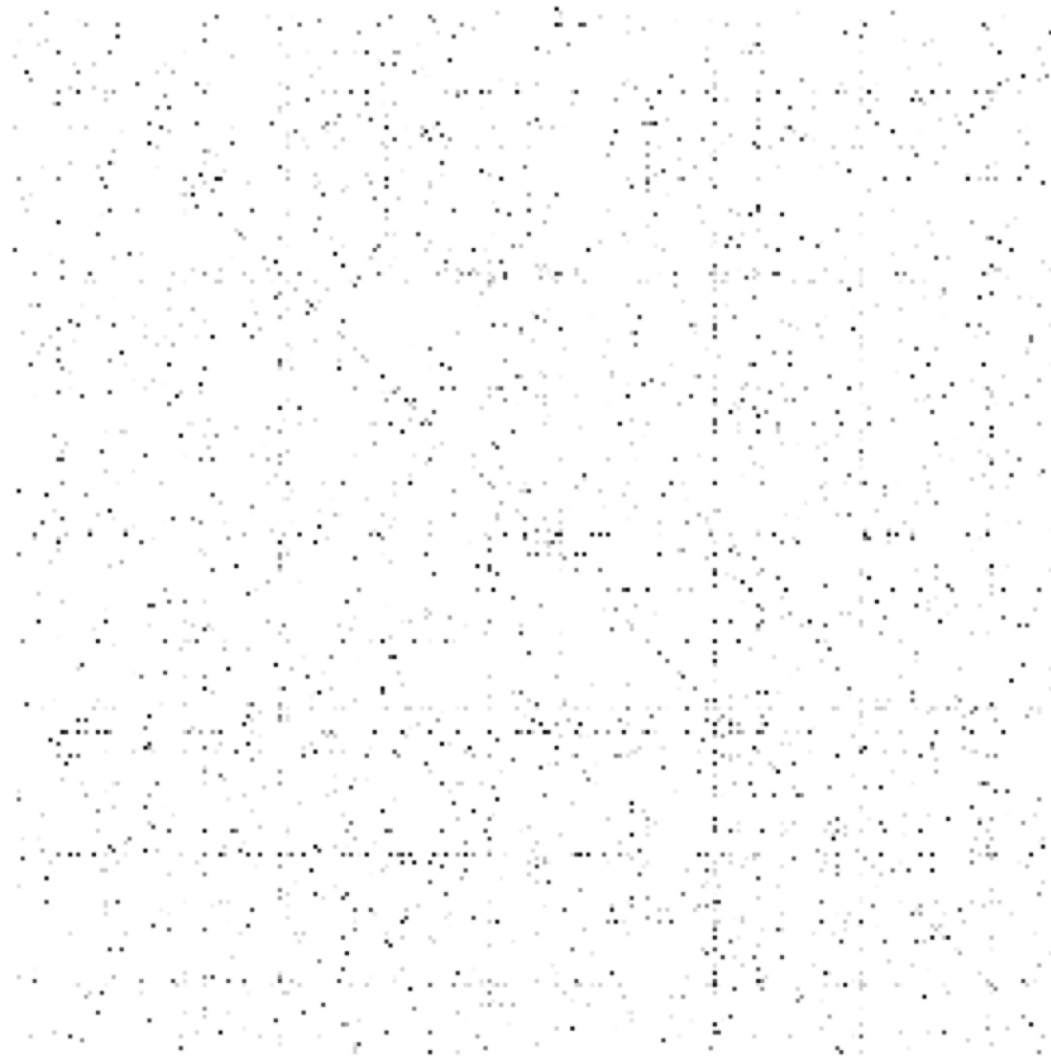
$$A_{ii} = 0$$

$$k_i^{out} = \sum_{j=1}^N A_{ij}$$

$$k_j^{in} = \sum_{i=1}^N A_{ij}$$

$$L = \sum_{i=1}^N k_i^{in} = \sum_{j=1}^N k_j^{out} = \sum_{i,j} A_{ij}$$

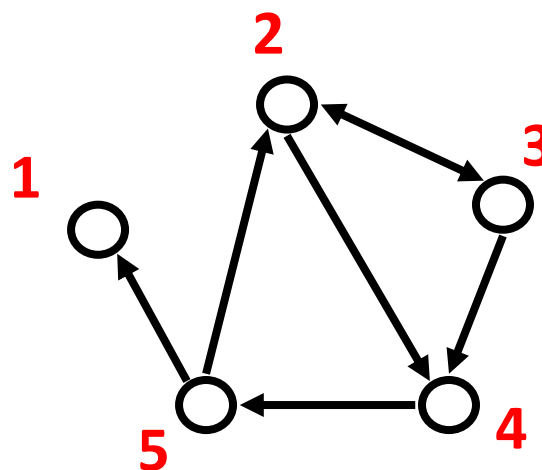
Adjacency Matrices are Sparse



Representing Graphs: Edge list

- Represent graph as a set of edges:

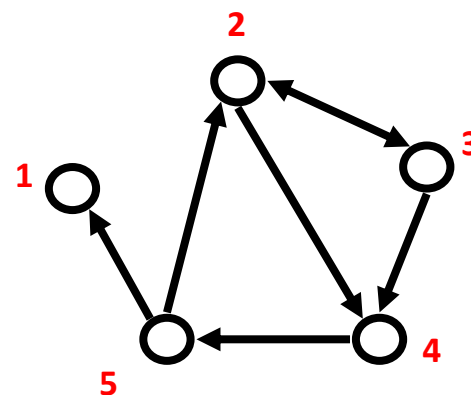
- (2, 3)
- (2, 4)
- (3, 2)
- (3, 4)
- (4, 5)
- (5, 2)
- (5, 1)



Representing Graphs: Adjacency list

■ Adjacency list:

- Easier to work with if network is
 - Large
 - Sparse
- Allows us to quickly retrieve all neighbors of a given node
 - 1:
 - 2: 3, 4
 - 3: 2, 4
 - 4: 5
 - 5: 1, 2



Networks are Sparse Graphs

Most real-world networks are **sparse**

$$E \ll E_{\max} \text{ (or } \bar{k} \ll N-1)$$

WWW (Stanford-Berkeley):	$N=319,717$	$\langle k \rangle=9.65$
Social networks (LinkedIn):	$N=6,946,668$	$\langle k \rangle=8.87$
Communication (MSN IM):	$N=242,720,596$	$\langle k \rangle=11.1$
Coauthorships (DBLP):	$N=317,080$	$\langle k \rangle=6.62$
Internet (AS-Skitter):	$N=1,719,037$	$\langle k \rangle=14.91$
Roads (California):	$N=1,957,027$	$\langle k \rangle=2.82$
Proteins (S. Cerevisiae):	$N=1,870$	$\langle k \rangle=2.39$

(Source: Leskovec et al., *Internet Mathematics*, 2009)

Consequence: Adjacency matrix is filled with zeros!

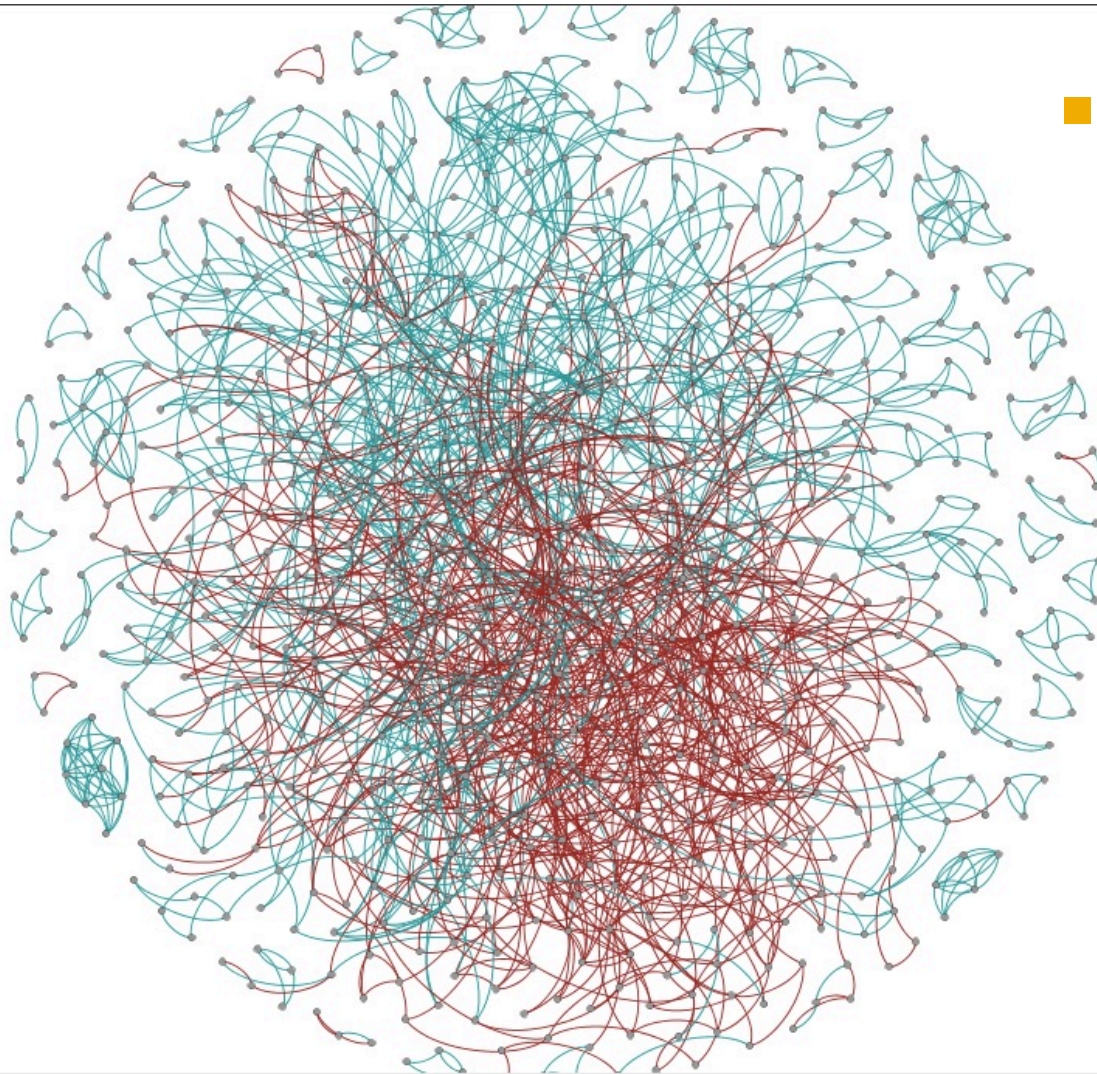
(**Density of the matrix (E/N^2):** WWW= 1.51×10^{-5} , MSN IM = 2.27×10^{-8})

Edge Attributes

Possible options:

- Weight (e.g. frequency of communication)
- Ranking (best friend, second best friend...)
- Type (friend, relative, co-worker)
- Sign: Friend vs. Foe, Trust vs. Distrust
- Properties depending on the structure of the rest of the graph: number of common friends

Positive and Negative Weights



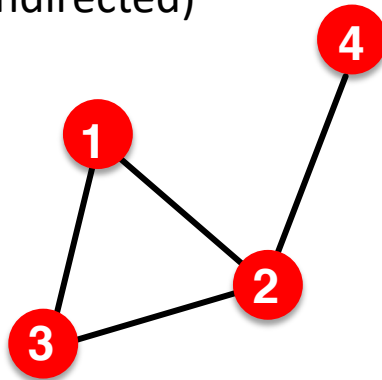
- **One person trusting/distrusting another**
 - Research challenge: How does one ‘propagate’ negative feelings in a social network? Is my enemy’s enemy my friend?

sample of positive & negative ratings from Epinions network

More Types of Graphs

■ Unweighted

(undirected)



$$A_{ij} = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

$$A_{ii} = 0$$

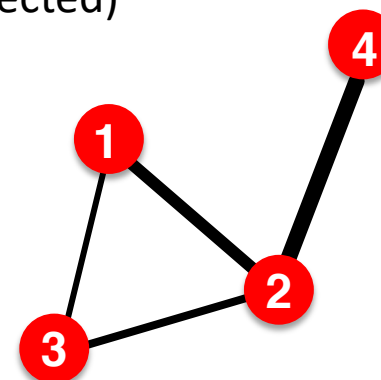
$$A_{ij} = A_{ji}$$

$$E = \frac{1}{2} \sum_{i,j=1}^N A_{ij} \quad \bar{k} = \frac{2E}{N}$$

Examples: Friendship, Hyperlink

■ Weighted

(undirected)



$$A_{ij} = \begin{pmatrix} 0 & 2 & 0.5 & 0 \\ 2 & 0 & 1 & 4 \\ 0.5 & 1 & 0 & 0 \\ 0 & 4 & 0 & 0 \end{pmatrix}$$

$$A_{ii} = 0$$

$$A_{ij} = A_{ji}$$

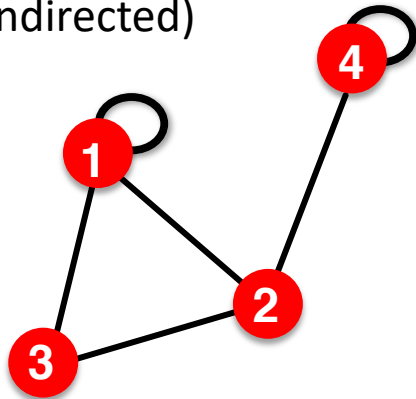
$$E = \frac{1}{2} \sum_{i,j=1}^N \text{nonzero}(A_{ij}) \quad \bar{k} = \frac{2E}{N}$$

Examples: Collaboration, Internet, Roads

More Types of Graphs

■ Self-edges (self-loops)

(undirected)



$$A_{ij} = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$

$$A_{ii} \neq 0$$

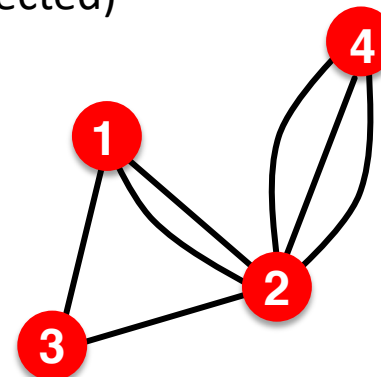
$$A_{ij} = A_{ji}$$

$$E = \frac{1}{2} \sum_{i,j=1, i \neq j}^N A_{ij} + \sum_{i=1}^N A_{ii}$$

Examples: Proteins, Hyperlinks

■ Multigraph

(undirected)



$$A_{ij} = \begin{pmatrix} 0 & 2 & 1 & 0 \\ 2 & 0 & 1 & 3 \\ 1 & 1 & 0 & 0 \\ 0 & 3 & 0 & 0 \end{pmatrix}$$

$$A_{ii} = 0$$

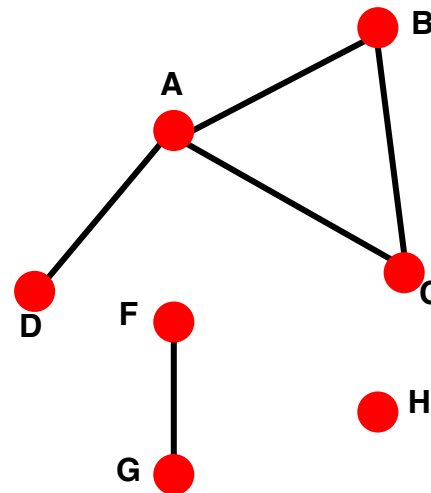
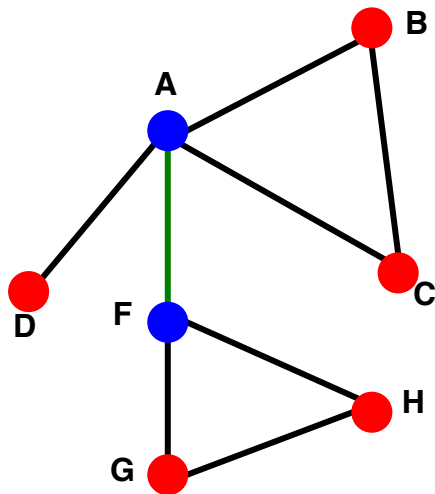
$$A_{ij} = A_{ji}$$

$$E = \frac{1}{2} \sum_{i,j=1}^N \text{nonzero}(A_{ij}) \quad \bar{k} = \frac{2E}{N}$$

Examples: Communication, Collaboration

Connectivity of Undirected Graphs

- **Connected (undirected) graph:**
 - Any two vertices can be joined by a path
- A disconnected graph is made up by two or more connected components



Largest Component:
Giant Component

Isolated node (node H)

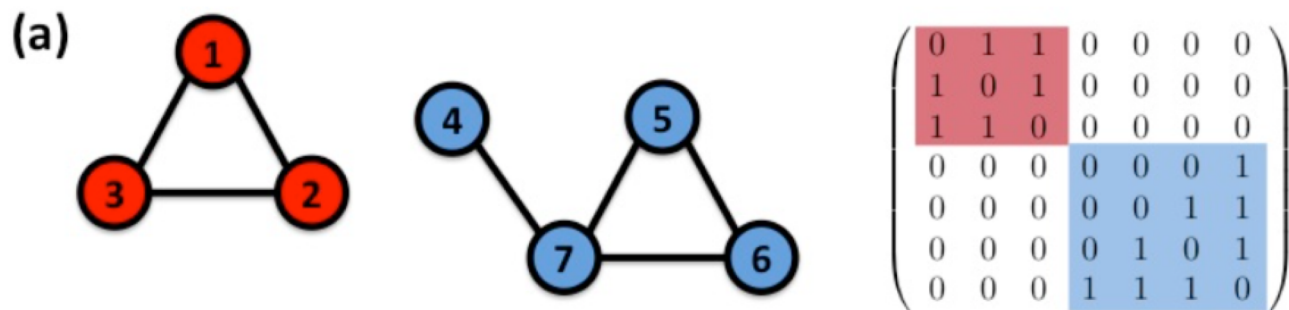
Bridge edge: If we erase the **edge**, the graph becomes disconnected

Articulation node: If we erase the **node**, the graph becomes disconnected

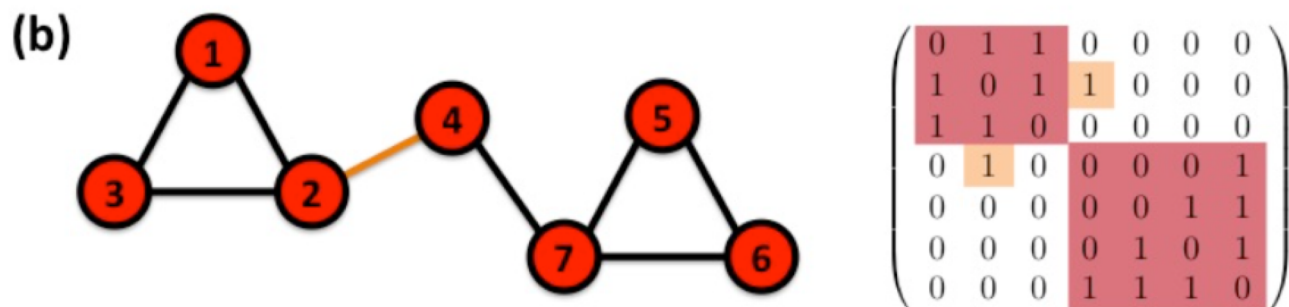
Connectivity: Example

- The adjacency matrix of a network with several components can be written in a block-diagonal form, so that nonzero elements are confined to squares, with all other elements being zero:

Disconnected

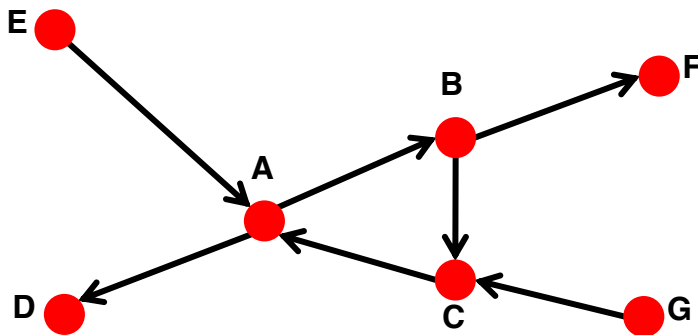


Connected



Connectivity of Directed Graphs

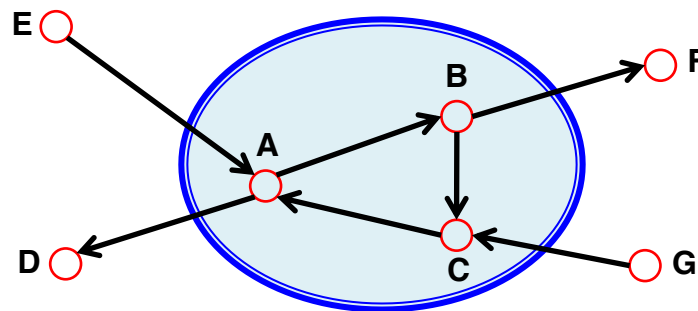
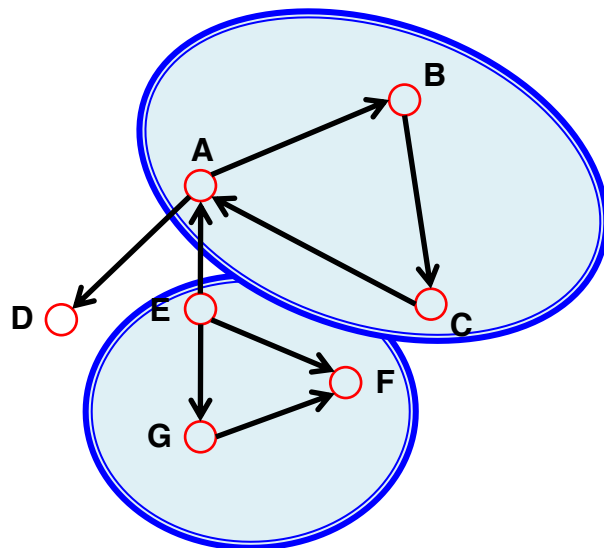
- **Strongly connected directed graph**
 - has a path from each node to every other node and vice versa (e.g., A-B path and B-A path)
- **Weakly connected directed graph**
 - is connected if we disregard the edge directions



Graph on the left is connected but not strongly connected (e.g., there is no way to get from F to G by following the edge directions).

Connectivity of Directed Graphs

- **Strongly connected components (SCCs)** can be identified, but not every node is part of a nontrivial strongly connected component.



In-component: nodes that can reach the SCC,

Out-component: nodes that can be reached from the SCC.

Network Representations

Email network >> directed multigraph with self-edges

Facebook friendships >> undirected, unweighted

Citation networks >> unweighted, directed, acyclic

Collaboration networks >> undirected multigraph or weighted graph

Mobile phone calls >> directed, (weighted?) multigraph

Protein Interactions >> undirected, unweighted with self-interactions